CSA04-Operating Systems

TASK 12 – Re-evaluate and Adapt the Design

A. Impact Analysis

The hospital introduces remote monitoring devices in patients' homes and moves patient records to a cloud service shared by three partner hospitals. This changes the original operating-system design by introducing internet communication, cloud computing and distributed data management.

Affected Operating-System Types, Services and Protection

Area	Original Design	Changed Requirement
Patient Monitoring	Local RTOS	Embedded RTOS with internet connectivity
Database	Local Linux server	Cloud-based distributed database
Communication	Hospital network	Secure internet communication
Security	Local authentication	Encryption and multi-factor authentication
Data Management	Local storage	Cloud storage and synchronization
Protection	Memory and access protection	Network security and access control

Impact: The system now requires stronger network security, remote access management, cloud storage and reliable communication between hospitals and home devices.

B. Evaluation of the Original Design

The original design provides good performance and security within hospitals, but some modifications are necessary to support remote monitoring and cloud records.

1. Response Time

- Local patient monitors provide fast responses.
- Internet communication may introduce network delays.
- Emergency alerts should be processed locally without waiting for cloud responses.

2. Reliability

- Cloud storage allows access to records from different hospitals.
- Internet failure may interrupt communication.
- Local buffering is required to store readings temporarily.

3. Security

- Remote devices may face unauthorized access and cyberattacks.
- Patient records require encryption during transmission and storage.
- Authentication and access control are necessary.

Evaluation: The original design is partially suitable, but it cannot fully support remote monitoring and shared cloud records without additional security and distributed-system services.

C. Adaptation or Redesign

I recommend modifying the original design by introducing embedded real-time systems, secure communication and cloud-based distributed storage.

Revised System Architecture

Remote Patient Monitoring Devices

Sensors • Embedded RTOS • Local alerts

Secure Internet Communication

Encryption • Authentication • TLS

Cloud Platform

Data processing • Distributed database • Backup

Hospital 1

Records

Hospital 2

Records

Hospital 3

Records

Revised Plan:

Component	**Recommended Solution**
Home Devices	Embedded RTOS
Local Emergency Alerts	Interrupt-driven processing
Data Transfer	Secure internet communication
Patient Records	Cloud-based distributed database

Data Storage	Cloud storage with backups
Security	Encryption, authentication and access control
Hospital Access	Secure web applications

D. Evidence-Based Defence

The modified design is better because it supports remote monitoring and centralized patient records while maintaining local emergency response capabilities.

Supporting Evidence:

1. Interrupt-driven processing provides faster sensor response than continuous polling.
2. DMA reduces CPU overhead when transferring large MRI images.
3. Caching improves the speed of frequently accessed patient records.
4. Cloud storage allows all three hospitals to access shared patient information.
5. Encryption, authentication and backups improve data security and reliability.

Trade-offs

Advantage	**Limitation**
Remote patient monitoring	Depends on internet connectivity
Centralized cloud records	Cloud service failure risk
Easy data sharing	Requires strong access control
Scalable storage	Additional cloud costs
Centralized management	Network latency

Final Evaluation: I recommend MODIFYING the original design rather than completely redesigning it. The existing RTOS and Linux server architecture can be retained, while embedded home-monitoring devices, secure internet communication and cloud-based distributed databases are added. This provides scalability, security and reliability for all three hospitals.

Conclusion

The modified operating-system design supports both hospital-based and home-based patient monitoring. It combines real-time processing, secure cloud storage, distributed data sharing and backup mechanisms. Therefore, the adapted design is more suitable for the hospital's new requirements while maintaining the important features of the original system.